

VATSAL THAKOR

Mumbai, India ♦ [LinkedIn](#) ♦ [Mail](#) ♦ [GitHub](#)

EDUCATION

PhD

CSIF department, IIT Bombay

July 2026 - Present

Relevant coursework: Materials Processing in Semiconductor Technology, Structural Characterization of Materials, Compound Semiconductor Materials and Devices

B.Tech in Electronics and Communication Engineering

Nirma University, Ahmedabad

Sept 2022 - May 2026 — CGPA: 7.88/10

Relevant coursework: Electronic Devices and Circuits, VLSI Design, Analog CMOS IC Design, Testing and Verification of Digital Circuits, FPGA-Based System Design, Digital Logic Design, Analog Electronics, Machine Learning

RESEARCH INTERESTS

Semiconductor Materials and Devices, III Nitride LEDs, Photonics, Advanced Memory Devices

SKILLS

Hardware Description Languages: Verilog HDL, SystemVerilog

Programming: Python, C, Embedded C, Assembly

EDA Tools: Cadence Virtuoso, Synopsys Sentaurus Topography 3D

FPGA Tools: Xilinx Vivado, Quartus, ModelSim

Technical Software: MATLAB, Microwind, Keil uVision

Mechanical CAD: SolidWorks, Creo Parametric

RESEARCH EXPERIENCE

Research Intern

January 2025 – April 2026

Indian Institute of Technology Gandhinagar (IITGN)

- Developed a 3D process simulation framework for plasma-based High Aspect Ratio (HAR) etching of ONO (Oxide–Nitride–Oxide) stacks used in 3D NAND memory structures.
- Implemented and calibrated Ar/C₄F₈/O₂ plasma chemistries in Synopsys Sentaurus Topography 3D to model polymer deposition, ion-enhanced etching, and radical transport.
- Analyzed radical angular distribution, reflection probability, reflection exponent, nucleation and growth probabilities to study polymer build-up, trench bowing, and distortion mechanisms.
- Optimized plasma process parameters to reduce polymer clogging at trench entrances and improve sidewall protection in high-aspect-ratio structures.
- Developed automated scripts to extract polymer thickness and volume metrics from Sentaurus TDR outputs for quantitative profile analysis.

PROJECTS

CMOS Operational Amplifier Design and Performance Analysis using Cadence Virtuoso

- Designed and simulated a CMOS operational amplifier in Cadence Virtuoso, analyzing key performance metrics, including Slew Rate, CMRR, PSRR, and GBW.
- Performed DC, AC, and transient simulations to extract and validate op-amp characteristics under various operating conditions.
- Optimized the transistor-level design for improved frequency response and stability, ensuring high performance and accurate analog behavior.

Raster-to-Vector Image Conversion using MATLAB Image Processing

- Developed a MATLAB-based image processing system for vectorizing raster images with adjustable parameters such as edge threshold, color quantization, and smoothing level.
- Implemented efficient matrix and vector operations to enhance processing speed and scalability for high-resolution images.
- Validated results through visualization and parameter tuning to achieve optimal edge accuracy and noise reduction across diverse image types.

FPGA-Based Morse Code Generator using Verilog HDL

- Designed a Morse code generator in Verilog to convert alphanumeric inputs into timed dot and dash sequences using a finite-state machine and lookup table.
- Simulated and verified functionality through testbenches before synthesizing and implementing it on an FPGA board.
- Validated timing accuracy and output through LED and buzzer signals for various input messages.

FSM-Based Secure Door Lock System with Three-Failure Alarm using Verilog HDL

- Designed a password-based door lock system in Verilog using a finite-state machine for secure access control.
- Simulated and tested functionality in Verilog before deploying on FPGA, ensuring reliable response and reset operation.
- Implemented a three-failure intrusion alarm mechanism triggered after consecutive incorrect password attempts.

UART-Based Serial Communication between Two 8051 Microcontrollers

- Developed a reliable serial communication protocol between two 8051 microcontrollers, enabling reliable UART-based data transmission and reception.
- Configured Timer 1 for baud rate generation, ensuring stable 9600 bps communication between transmitter and receiver microcontrollers.
- Implemented logic to compare received data and trigger specific hardware actions based on the input, demonstrating efficient data handling and control.